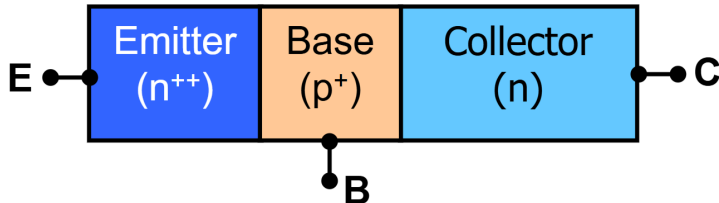
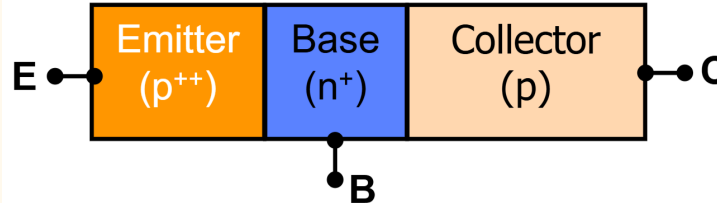
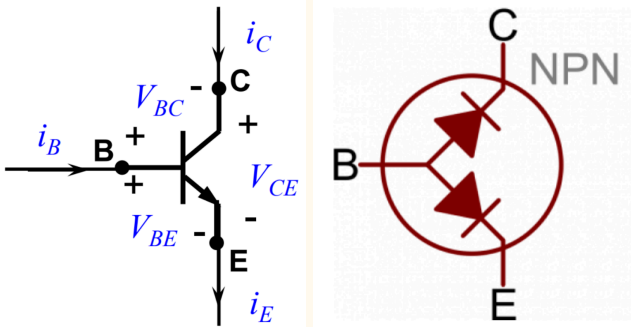
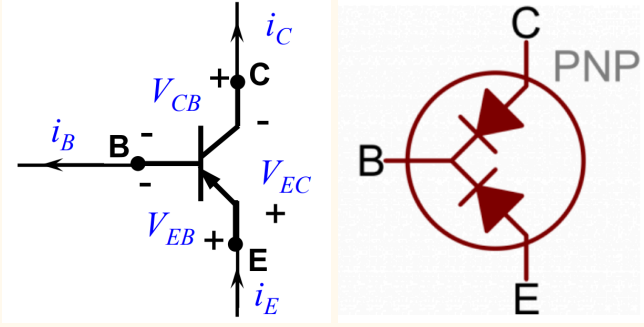


BJT Summary

Operating Modes

npn	pnp																																								
Active region of npn BJT: 	Active region of pnp BJT: 																																								
																																									
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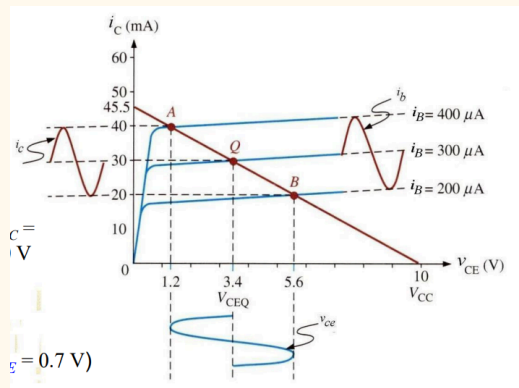
Forward Active IV Characteristics

Without Early Effect	With Early Effect ($V_A \sim 100\text{-}200\text{V}$)
$i_C = I_S e^{v_{BE}/V_T}$ $i_B = \frac{I_S}{\beta} e^{v_{BE}/V_T}$	$i_C = I_S e^{v_{BE}/V_T} \left(1 + \frac{v_{CE}}{V_A}\right)$ $i_B = \frac{I_S}{\beta} e^{v_{BE}/V_T} \left(1 + \frac{v_{CE}}{V_A}\right)$

$i_E = i_C + i_B \approx i_C$	$i_C = \beta i_B$	$i_E = (\beta + 1) i_B$	$i_E = \left(\frac{1}{\beta} + 1\right) i_C$
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$\alpha = \beta/(1+\beta)$

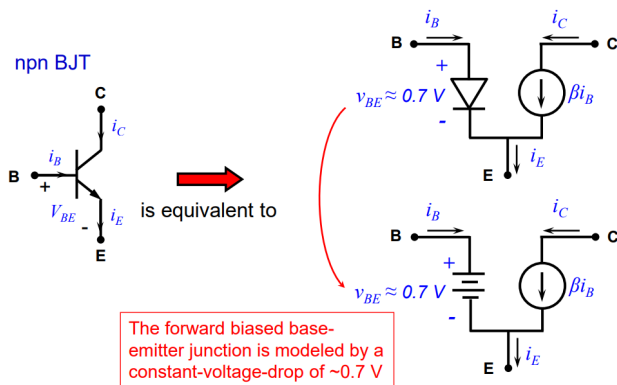
Modelling



DC

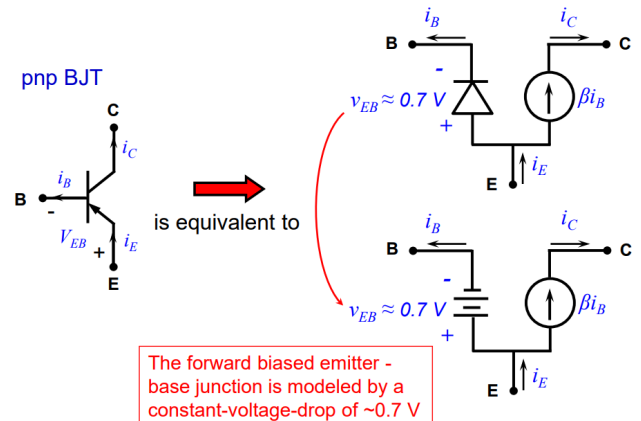
npn

As the emitter-base junction is forward biased, $V_{BE} \approx 0.7$ V, it can be modeled by a constant voltage source (using the constant-voltage-drop model for a forward biased pn-junction).



pnp

Large-signal model of a pnp BJT: polarities of voltages and directions of currents are opposite to those of an npn BJT.



- direction of current source and voltage drop is opposite

Thevenin Equivalent Method

- Check npn or pnp
- Assumption:
 - BJT is under forward active mode $\Rightarrow I_C = \beta I_B$
- Model base biasing circuit by Thevenin equivalent
- With the Thevenin equivalent, find base current
- With base current and β find collector current
- With collector current find emitter current and collector voltage
- Check operating mode to verify assumption
 - for npn: $V_{BE} > 0.7$ V, emitter-base is forward biased & $V_{BC} < 0$ V, base-collector is reverse biased
 - for pnp: $V_{EB} > 0.7$ V, emitter-base is forward biased & $V_{CB} < 0$ V, base-collector is reverse biased

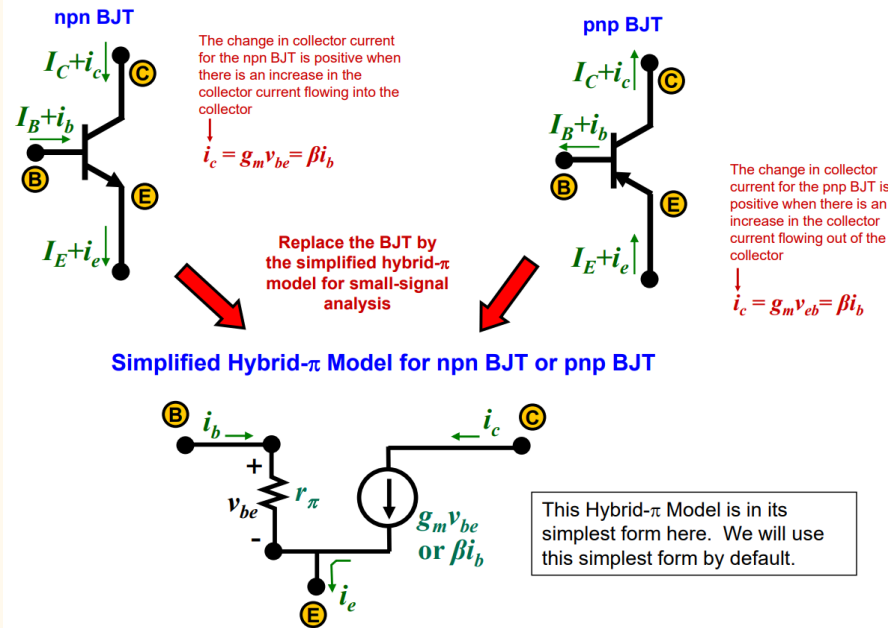
Voltage divider method

- Check npn or pnp
- Assumption:
 - BJT is under forward active mode $\Rightarrow I_C = \beta I_B$
 - Biassing circuit current $\gg$ Base current $\Rightarrow$ base voltage can be easily found
- From potential divider, find base voltage
- From base voltage and 0.7 V drop, find emitter voltage
- From emitter voltage and emitter resistor, find emitter current
- From emitter current and β , find collector current and base current
- From collector current and collector resistor find collector voltage
- Check base current and biassing circuit current to verify assumption 2
 - if biassing circuit current is not 10x of base current, redo analysis with Thevenin method
- Check operating mode to verify assumption 1

- a. for npn: $V_{BE} > 0.7$ V, emitter-base is forward biased & $V_{BC} < 0$ V, base-collector is reverse biased
b. for pnp: $V_{EB} > 0.7$ V, emitter-base is forward biased & $V_{CB} < 0$ V, base-collector is reverse biased

AC

Simplified Hybrid- π Model



$$r_{\pi} = r_i$$

EE2027 used r_{π}
EE5507 used r_i

$$\beta = g_m r_{\pi}$$

- ✧ Its easy to remember this
- ✧ Wherever the input current is, it goes into r_{π} and generate a voltage
- ✧ Output current is depending on the voltage via g_m
- ✧ So, the overall current gain is simply $g_m r_{\pi}$

No Early Effect

$$i_c = I_S e^{v_{BE}/V_T}$$

Parameters

$$g_m = \text{Transconductance}$$

$$g_m = \frac{I_C}{V_T}$$

$$r_{\pi} = \text{input resistance}$$

$$r_{\pi} = \frac{\beta V_T}{I_C}$$

$$\beta = g_m r_{\pi}$$

$$r_{\pi} = \frac{\beta}{g_m}$$

$$g_m = \frac{\beta}{r_{\pi}}$$

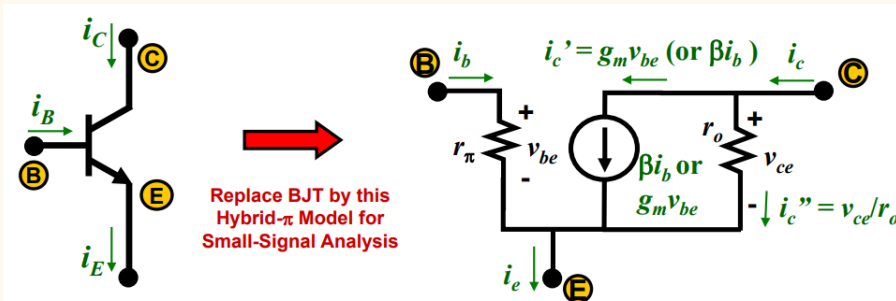
Relationships

$$i_c = g_m v_{be}$$

$$v_{be} = r_{\pi} i_b$$

$$i_c = \beta i_b$$

Full Hybrid- π Model with Early Effect



Early Effect

$$i_c = I_S e^{v_{BE}/V_T} \left(1 + \frac{v_{CE}}{V_A}\right)$$

Additional Parameter

$$r_o = \text{output resistance}$$

$$r_o = \frac{V_A}{I_C}$$

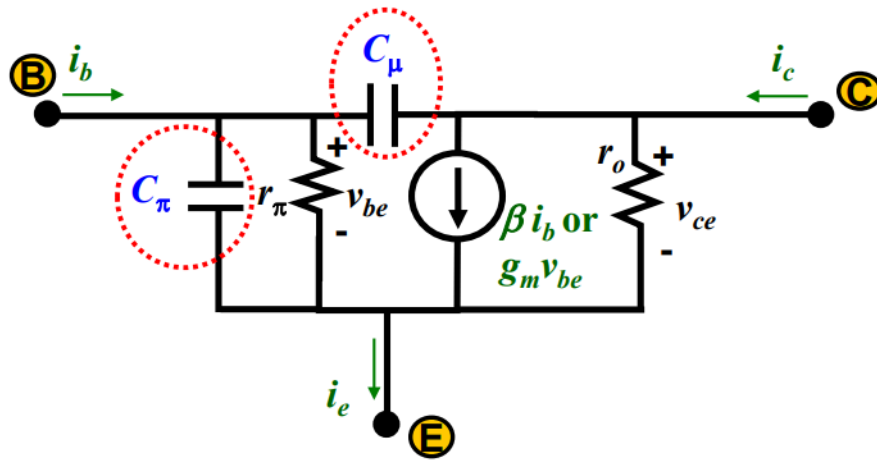
$$r_o = \infty \text{ if } V_A \text{ not given}$$

Relationships

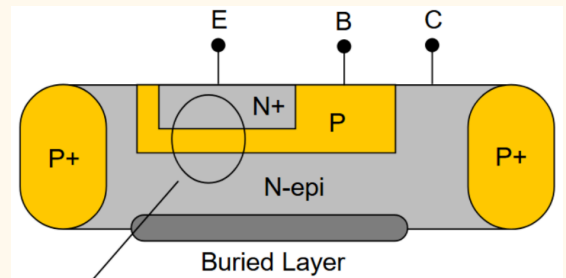
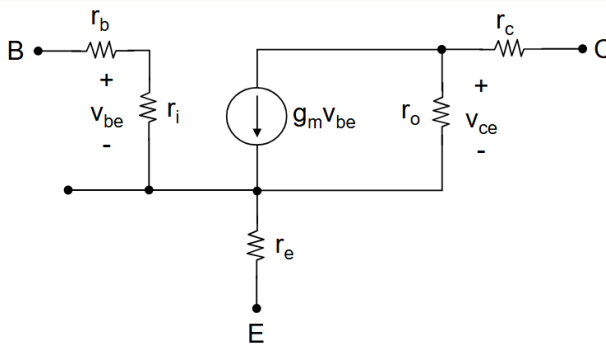
$$i_c = g_m v_{be} + \frac{1}{r_o} v_{ce}$$

$$i_c = \beta i_b + \frac{1}{r_o} v_{ce}$$

Full Hybrid- π Model at High Frequency (EE2027)



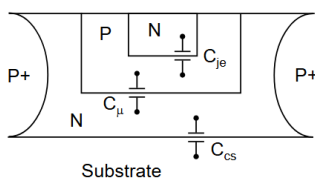
Full Hybrid- π Model at Low Frequency (EE5507)



r_e is small, because the emitter is on top and lightly doped
 r_c is non negligible without the buried layer
 r_b can be minimised with fingers, like R_g in MOSFET, see [here](#)

Full Hybrid- π Model at High Frequency (EE5507)

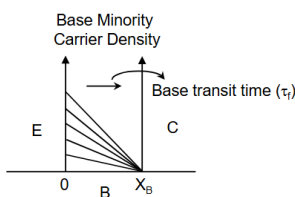
- Junction Capacitance (C_{je} , C_{μ} , C_{cs})



$$C_j = \frac{C_{j0}}{\sqrt[m]{1 + \frac{V_{Bias}}{\phi_{bi}}}} \quad m \approx \frac{1}{3} \sim \frac{1}{2}$$

C_{je} : forward bias
 C_{μ} , C_{cs} : reverse bias

- Base Charging Capacitance (C_b)



$$\Delta Q_n = i_c \cdot \tau_F$$

$$\Delta Q_n = C_b \cdot v_{be}$$

$$\Delta Q_n = \Delta Q_h \quad \leftarrow \text{Charge Neutrality}$$

$$\Rightarrow C_b = \tau_F \cdot \frac{i_c}{v_{be}} = \tau_F \cdot g_m$$

$$\approx 250 \text{ ps} \cdot \frac{1}{26 \Omega} = 10 \text{ pF}$$

Diffusion = Base Charging capacitance
 $\gg$ In NPN, base is forward bias then a lot of electrons injected into the base then get swept to the collector.

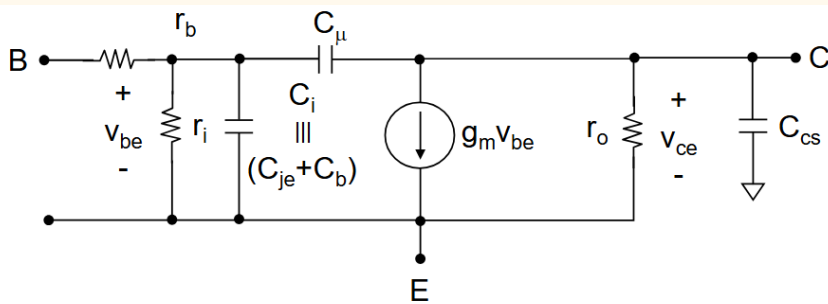
$\gg$ Base terminal will have +ve charge, then the negative electron is in the base region.

$\gg$ Turning BJT off will not be instantly, it takes time for electrons to move from the base region and get collected by the collector.

$\gg$ This is called transit time, τ_F
 $\gg$ Can deduce the diffusion capacitance from transit time

$\gg$ Dominant factor for the BJT speed

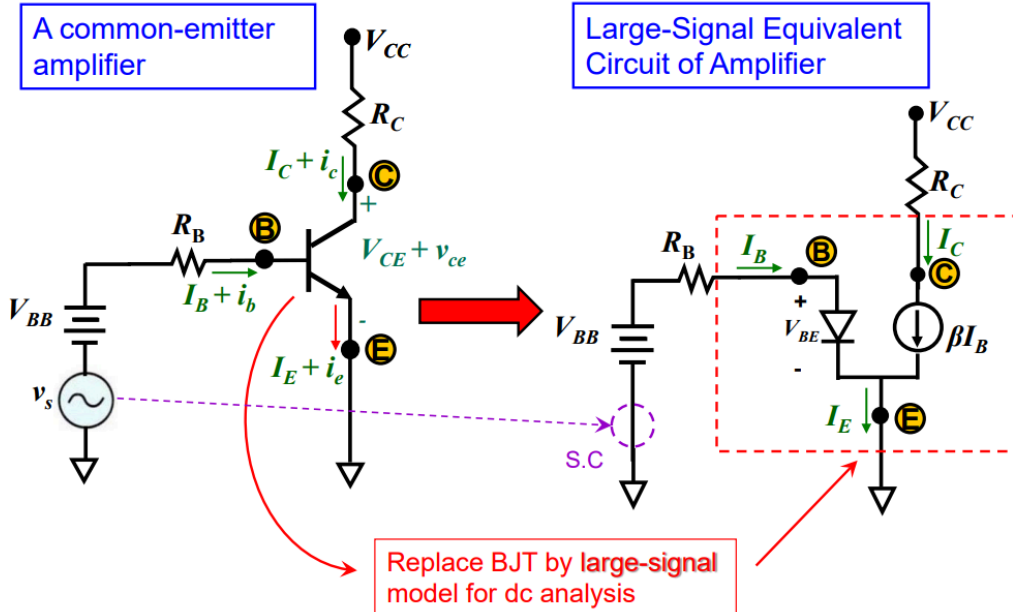
make the E to C width thinner = base width thinner, then BJT faster



$$C_b \gg C_{je}$$

Full Analysis

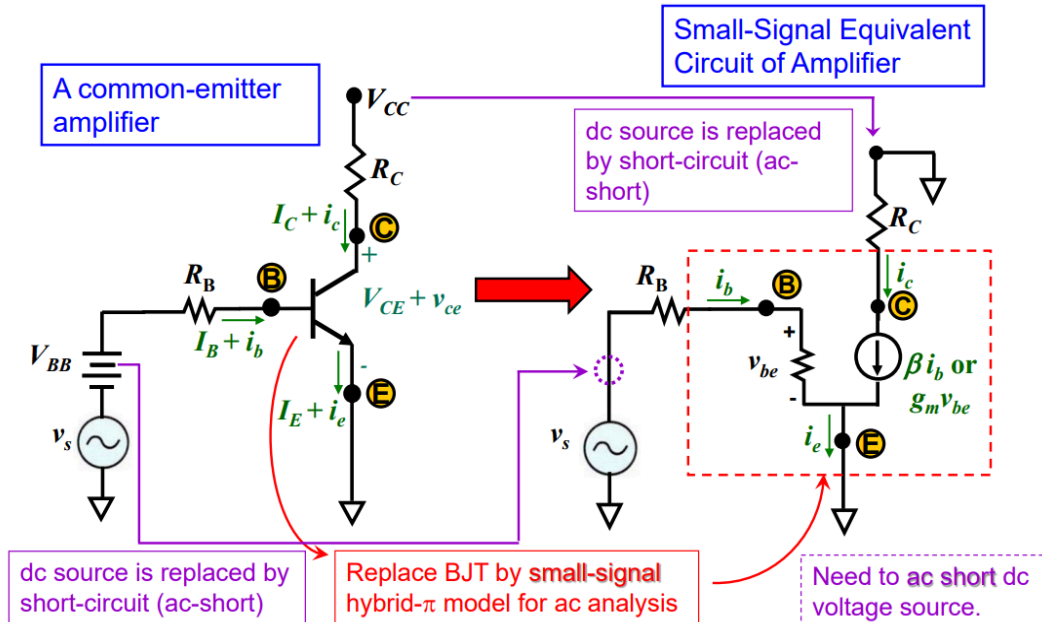
- The dc bias point of the amplifier can be determined using the **large-signal** equivalent circuit, in the **absence** of the ac (small signal) source, v_s .



Turn off AC source and perform DC analysis to obtain DC biasing point

I_C is used to find the small signal parameters

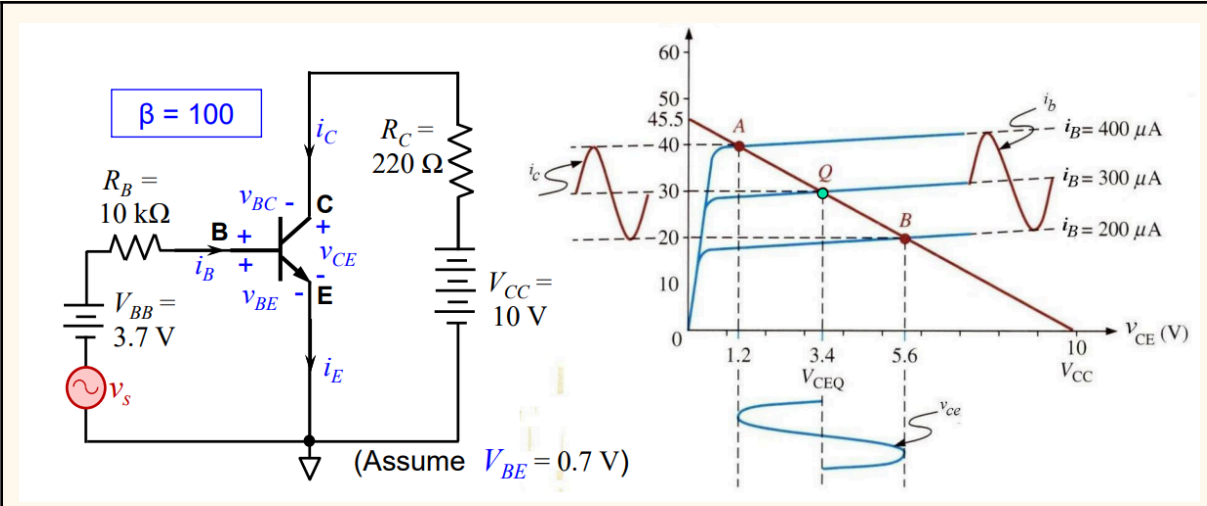
- The small-signal ac analysis of the amplifier is carried out using the **small-signal** equivalent circuit, and dc voltage sources need to be ac short.



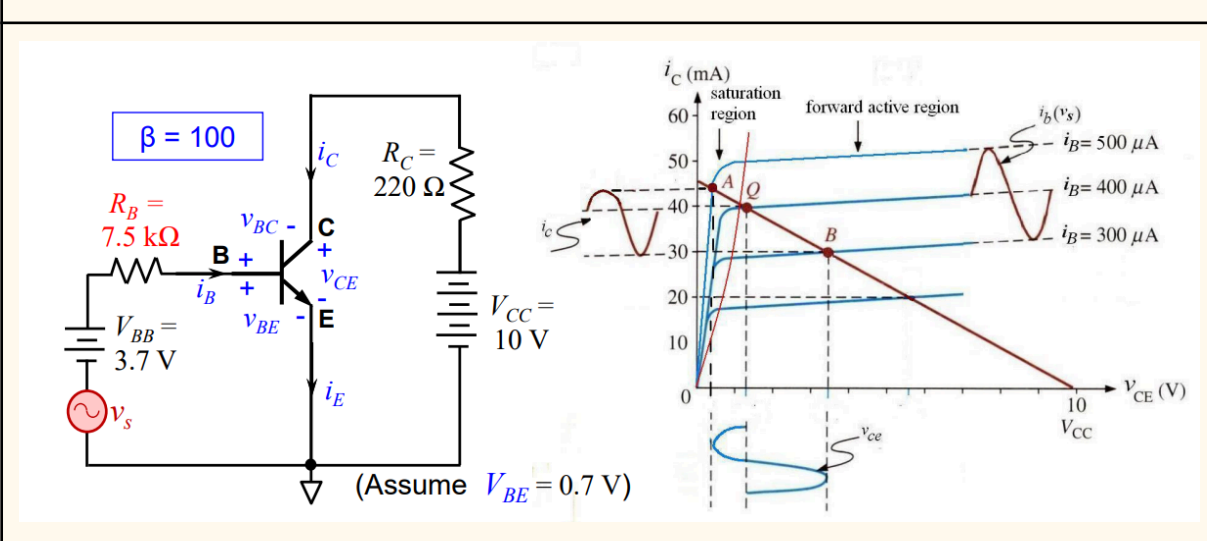
Turn off DC source, this includes the V_{CC}

Perform AC analysis

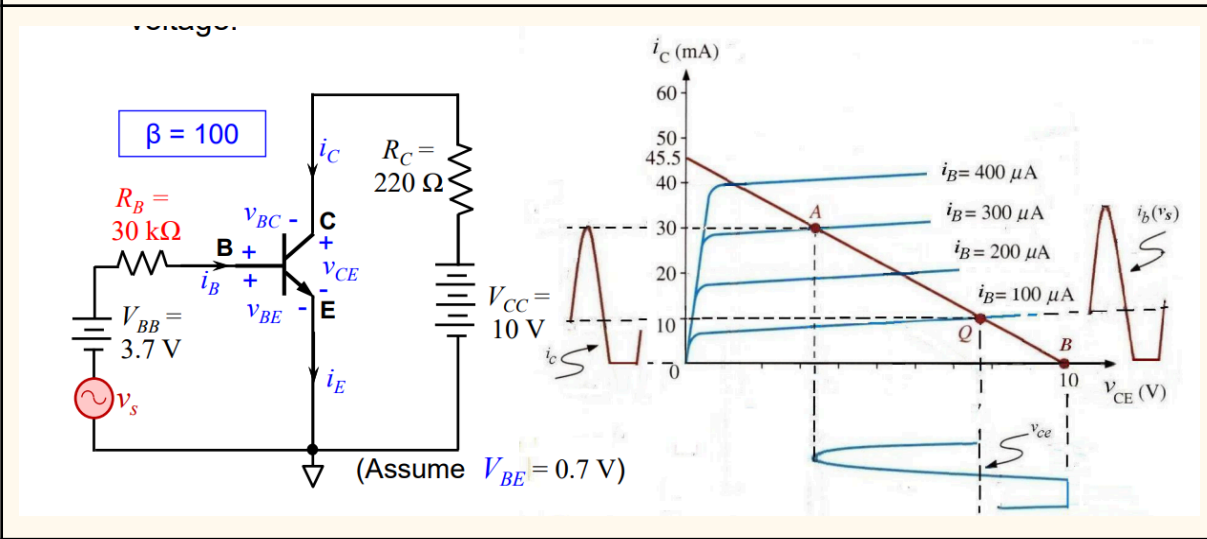
Art of Amplifier Biasing



Correct Biasing
No distortion from clipping or non linearity



Biasing Too High
Distortion from non linearity due to entering saturation region



Biasing Too Low
Distortion from clipping due to entering cut-off region